

DEDAN OKWARE

Senior Software Engineer · Solution Architect — Python/Django · React · AI Systems · Secure Beneficiary Data

Deep front-end specialisation carried on senior engineering and architecture practice

softengdedan@gmail.com · Nairobi, Kenya · github.com/okwareddevnest · linkedin.com/in/softcysec-dedan-okware · my-portfolio-crypt.vercel.app

PROFESSIONAL SUMMARY

Software engineer and architect working at the intersection of the three things HOPE depends on: Python/Django service architecture, applied AI systems, and the handling of confidential personal data inside regulated financial workflows. Currently Lead Front End Engineer at Fingo Africa, owning architecture across a cross-border payments and stablecoin platform — wallets, on/off-ramp rails, transfer flows and KYC onboarding, the same operational shape as a cash-transfer management information system moving payment lists to financial service providers and reconciling them afterwards. Previously built AI-powered document verification and evidence-matching systems for a UK immigration platform, and designed tamper-evident storage for the records those decisions rested on. Active open-source maintainer with tools in daily use by 3,500+ developers across 40+ countries. Depth runs in two directions at once: front-end engineering to production specialist standard across TypeScript, React, Next.js, Vue and Flutter, and senior software engineering across architecture, backend services, data design and technical leadership of delivery teams — most recently as Technical Lead Software Engineer at Power Learn Project Africa, leading the setup and revamp of core systems through an organisation-wide pivot. Based in Nairobi, available to the duty station without relocation.

CORE COMPETENCIES

Languages · Python · TypeScript · JavaScript (ES6+) · Golang · Rust · Motoko · Solidity · SQL

Backend & Architecture · Django · FastAPI · Node.js · Express.js · REST · GraphQL · gRPC · Microservices · Systems Integration · Event-Driven & Queue-Based Design · API Contract Design

Data · PostgreSQL · MySQL · MongoDB · Redis · Pandas · NumPy · Schema Design · Data Retention & Archiving · Migration Safety

AI & Machine Learning · LLM Integration · Autonomous Agent Orchestration · Agent Tooling & Skill Interfaces · LangChain · TensorFlow · PyTorch · Scikit-learn · Document Classification & Verification · Evidence Matching & Anomaly Heuristics

Front End · React · Next.js · Vue.js · Flutter · TypeScript · Component Architecture · Design Systems · Accessibility (WCAG)

Security & Data Protection · Applied Cryptography · Signature Verification · Tamper-Evident Storage · KYC / Identity Verification Flows · Secure Handling of Confidential Personal Data · Google Cybersecurity Professional Certificate

Platform & Delivery · Docker · Kubernetes · Azure · AWS · Nginx · Linux · Git · GitHub Actions · CI/CD · Release Management · n8n Workflow Automation

Ways of Working · Agile Delivery · Multi-Stakeholder Coordination · Technical Standards & Governance · Code Review at Scale · Vendor and Contributor Oversight · Technical Mentorship · Documentation

PROFESSIONAL EXPERIENCE

Lead Front End Engineer — Fingo Africa

Feb 2026 – Present · Nairobi, Kenya · Hybrid, Full-time

- Own architecture and delivery across Fingo Global, a regulated cross-border payments and stablecoin platform: wallets, on/off-ramp interfaces, transfer flows and KYC/onboarding — beneficiary identity, payment instruction and reconciliation concerns in the same system.
- Define and enforce engineering standards, code review practice and testing frameworks across the engineering team, and own the deployment pipeline, CI/CD and release management for both web and mobile surfaces.
- Design identity and onboarding flows against compliance constraints, where an unverified record or a lost verification state is a regulatory incident rather than a cosmetic defect.
- Translate product specifications from the Stablecoin Product Lead and Engineering Lead into implementable technical designs, negotiating scope across product, design and backend engineering.

Blockchain Software Engineer (Contract) — Bonded · London, United Kingdom

Apr 2025 – Jul 2025 · Remote

- Built AI-powered evidence-matching algorithms that aligned and validated relationship documentation submitted by two parties across borders — automated document checking against a rules-and-scoring model, with human review on low-confidence outcomes.
- Delivered internal document-verification and case-communication systems in Flutter, streamlining manual review for a UK visa application platform handling sensitive personal records.
- Implemented Internet Computer smart contracts for tamper-evident storage of application evidence and verification records, giving each decision an auditable, immutable provenance trail.
- Worked across three time zones with a distributed cross-functional team, delivering a compliance-sensitive system to schedule.

Technical Lead Software Engineer — Power Learn Project Africa

Jan 2026 – May 2026 · Nairobi, Kenya · Consultancy

- Technical lead for the engineering function during the organisation's pivot into a new operating model, owning architecture decisions across backend services, front-end platforms and automation infrastructure.
- Led the setup and revamp of core internal systems: consolidated fragmented platforms, defined the migration path off legacy workflows, and sequenced the work so the organisation kept operating throughout.
- Set and enforced technical direction and engineering standards across the team, reviewing designs and implementations end to end — the same architecture, design framework and governance remit as deliverable 1 of these Terms of Reference.
- Delivered against a compressed timeline while organisational strategy was itself still moving, coordinating across programme, operations and external stakeholders.

Software Engineer & Instructor — Power Learn Project Africa

Oct 2024 – May 2026 · Nairobi, Kenya · Consultancy

- Extended a continent-scale learning platform with Golang backend services, Next.js and React front ends and a Strapi CMS, deployed in Docker on Scaleway infrastructure.
- Automated organisational standard operating procedures with a self-hosted n8n server and 15+ webhook integrations across internal systems, cutting manual operations by 60%.
- Performed 500+ code reviews and ran 100+ live engineering sessions, holding a consistent technical standard across a large and highly distributed contributor base — the same oversight discipline a vendor-delivered codebase requires.
- Trained and graduated 9,000+ engineers across Africa in full-stack development over 3 cohorts, and authored 15+ capstone projects adopted as the academy standard. Direct experience of delivery constraints in low-bandwidth, low-resource operating contexts across the continent.

Freelance Software Engineer

Apr 2023 – Mar 2025 · Kenya · Remote

- Architected and delivered 20+ production systems for clients across Africa and internationally, spanning full-stack web platforms, cross-platform mobile applications and decentralised applications.
- Built production APIs and microservices in Python, Node.js and Golang serving 50,000+ monthly users, owning design, data modelling, deployment and operation end to end.
- Sustained a 100% client satisfaction rate with 70% repeat engagement.

Open Source Developer & Maintainer

Jan 2023 – Present · Remote

- Maintain 10+ public repositories, triaging and merging 500+ issues and pull requests from external contributors under a published contribution standard.
- **Gitok** — Git productivity CLI with 35+ commands, adopted by 2,000+ developers across 40+ countries, with a fully automated semantic-versioned release pipeline.
- **U-Download** — cross-platform desktop application on a Rust/Tauri core with a React/TypeScript interface, trusted by 1,500+ users, packaged natively for Linux, Windows and macOS.
- **RSON** — a data serialisation format extending JSON with comments and richer types, with parsers in Rust, TypeScript and Python and a written language specification.

SELECTED WORK RELEVANT TO THIS ASSIGNMENT

OHMS 2.0 — Autonomous AI Agent Platform · Rust, TypeScript, React 19, Internet Computer, LLM integration, WebAssembly

A platform for composing autonomous agents from natural-language intent, with verifiable on-chain execution. Designed the agent-skill interface so capabilities are declared independently of any single model or agent runtime, and built the full React 19 + TypeScript composition surface. Directly analogous to agent-agnostic custom AI skills and ecosystem AI development support. **WCHL Blockchain Championship — 2nd Place National (Kenya) and 2nd Place Regional (Africa); Global Finalist.**

AI Evidence Matching & Document Checking — Bonded · *Python, AI/ML, Flutter, ICP*

Automated verification of paired identity and relationship documents for immigration casework: extraction, cross-party consistency checks, confidence scoring and escalation to human review, with tamper-evident storage of every artefact behind the decision. The closest available analogue to a document checker service, anomaly flagging and beneficiary record authentication.

Cross-Border Payment & KYC Platform — Fingo Global · *Flutter, Next.js, TypeScript*

Production interfaces for regulated value transfer: identity onboarding, transfer initiation, and the operational states that surround money movement — pending, failed, reconciled, disputed.

CERTIFICATIONS

- **Google Cybersecurity Professional Certificate** — Coursera, 2023
- **ICP Rust Smart Contract 101** — Dacade.org, 2024
- **ICP TypeScript Smart Contract 101** — Dacade.org, 2024

AWARDS

- **WCHL Blockchain Championship 2025** — 2nd Place, National Round (Kenya); 2nd Place, Regional Round (Africa); qualified to the Global Finale.

EDUCATION

Software Engineering & Computer Science — specialised in full-stack development, distributed systems and software architecture, with continuous professional development through certification and applied project work.

LANGUAGES

English — full professional proficiency.